

CS 2614: Computer Organization

Homework 8 Solutions

5.2. What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register?

Solution:

A direct address instruction needs two references to memory:

- (1) Read instruction,
- (2) Read operand.

An indirect address instruction needs three references to memory:

- (1) Read instruction,
- (2) Read effective address,
- (3) Read operand.

5.4. The following register transfers are to be executed in the system of Fig. 5-4. For each transfer, specify: (1) the binary value that must be applied to bus select inputs S_2 , S_1 , and S_0 ; (2) the register whose LD control input must be active (if any); (3) a memory read or write operation (if needed); and (4) the operation in the adder and logic circuit (if any).

- a. $AR \leftarrow PC$
- b. $IR \leftarrow M[AR]$
- c. $M[AR] \leftarrow TR$
- d. $AC \leftarrow DR$, $DR \leftarrow AC$ (done simultaneously)

Solution:

	(1) $S_2 S_1 S_0$	(2) <u>Load(LD)</u>	(3) <u>Memory</u>	(4) <u>Adder</u>
(a) $AR \leftarrow PC$	010 (PC)	AR	—	—
(b) $IR \leftarrow M[AR]$	111 (M)	IR	Read	—
(c) $M[AR] \leftarrow TR$	110 (TR)	—	Write	—
(d) $DR \leftarrow AC$ $AC \leftarrow DR$	100 (AC)	DR and AC	—	Transfer DR to AC

5.7. What are the two instructions needed in the basic computer in order to set the E flip-flop to 1?

Solution:

CLE Clear E
CME Complement E

5.8. Draw a timing diagram similar to Fig. 5-7 assuming that SC is cleared to 0 at time T_3 if control signal C_7 is active.

$C_7 T_3$: $SC \leftarrow 0$

C_7 is activated with the positive clock transition associated with T_1 .

Solution:

